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REVIEW

Cognitive behavioural therapy for depression in women with PCOS: systematic review and meta-analysis

BIOGRAPHY

Geranne Jiskoot works as a psychologist and obtained a PhD degree at the Department of Reproductive Medicine of the Erasmus MC. Her main research interests include lifestyle treatment in women with PCOS and the emotional aspects of premature ovarian insufficiency.

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KEY MESSAGE

Most psychological interventions applying cognitive behavioural therapy are effective in lowering depression scores in adult women with PCOS. An overall Cohen's *d* effect size of 1.02 was found in favour of CBT. However, these results should be interpreted with caution due to methodological differences and quality of the studies.

ABSTRACT

Polycystic ovary syndrome (PCOS) is a common endocrine disorder with physical and psychological complaints, especially high depression scores. Cognitive behavioural therapy (CBT) is the first-line psychological treatment for depression. The objective of this study was to examine the effect of different types of CBT interventions and the effects on depression scores in women with PCOS. A literature search was performed in six databases up to July 2020. Studies published in English, in which depression scores were compared between groups during a CBT intervention in women with PCOS, were included. A total of 4854 studies were identified, of which eight studies were included in the systematic review and five in the meta-analysis. CBT ranged from 8 to 52 weeks and involved between 8 and 20 sessions. An overall Cohen's *d* effect size of 1.02 (95% confidence interval 0.02–2.02) was found in favour of CBT compared with standard care. To conclude, most psychological interventions applying CBT are effective in lowering depression scores in women with PCOS. These results should be interpreted with caution due to methodological differences and quality of the studies. More clinical trials are needed to assess how many sessions of CBT are necessary to treat depression in women with PCOS.

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KEYWORDS

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INTRODUCTION

Polycystic ovary syndrome (PCOS) is a common endocrine disorder that affects 5–10% of women in their reproductive years (*Ehrmann, 2005*). Most women with PCOS experience one or more of the following physical symptoms in varying degrees: irregular menstrual periods, subfertility, hirsutism (excessive body hair growth), acne, obesity, insulin resistance and dyslipidaemia (*Laven et al., 2002*; *Rotterdam ESHRE/ASRM-Sponsored PCOS Consensus Workshop Group, 2004*). Besides these physical symptoms, many women with PCOS experience psychological symptoms such as depression (due to daily fatigue, sleep disturbances, appetite changes and diminished interest), anxiety, bulimia nervosa, and disordered eating behaviours. Moreover, they also report more often low health-related quality of life (QoL), sexual dissatisfaction and appear to have low self-esteem and a negative body image (*Annagur et al., 2014*; *Cesta et al., 2016*; *de Niet et al., 2010*; *Hollinrake et al., 2007*; *Lee et al., 2017*). In particular, depression and anxiety scores are significantly higher (*Veltman-Verhulst et al., 2012*) and remained consistently high across the lifespan in women with PCOS compared with a normal population (*Greenwood et al., 2019*). Cooney and colleagues found a median prevalence for depression of almost 37% among women with PCOS compared with 14% in controls (*Cooney et al., 2017*). Most women with PCOS score mild and moderate on the Beck Depression Inventory (BDI) with a mean BDI score of 12.7 (*Tan et al., 2008*). Despite the evidence that women with PCOS are at increased odds for depression and anxiety, there is no evidence supporting a single pathway for these increased odds (*Cooney and Dokras, 2017*). The presence of acne, hirsutism, insulin resistance, infertility due to oligo-ovulation and obesity seems to be associated with depressive feelings and lower QoL (*Hahn et al., 2005*; *Teede et al., 2018)*. Others found that all women with PCOS had difficulties in dealing with menstrual function, fertility and body image and found no difference between the depressed and non-depressed group (*Kerchner et al., 2009*). Based on qualitative research, women with PCOS perceive themselves as abnormal and less feminine due to hirsutism, menstrual dysfunction and subfertility (*Kitzinger and Willmott, 2002*). Therefore, researchers suggested that the chronic and complex nature of PCOS can result in high levels of stress (*Teede et al., 2010*). Another important factor that could be associated with mental health is the availability of information about the disorder for this group of women (*Avery and Braunack-Mayer, 2007*). A large international survey among women with PCOS revealed that only 3.4% of the women were satisfied with the emotional support and counselling they received after the PCOS diagnosis. Also, in more than 60% of the participants emotional support was not offered or discussed (*Gibson-Helm et al., 2017*).

According to the recent international PCOS guideline (*Teede et al., 2018*), risk factors and the severity of depressive as well as anxiety symptoms should be routinely screened for all women with PCOS. These international guidelines further suggest that depressive complaints in women with PCOS should be treated according to regional protocols (*Brennan et al., 2017*; *Teede et al., 2010*). In most countries, the first-line treatment for depression is cognitive behavioural therapy (CBT) or interpersonal therapy and behavioural activation. For maintenance treatment, CBT and mindfulness-based cognitive therapy are recommended (*Parikh et al., 2016*). CBT is based on the cognitive model: 'the way that individuals perceive a situation is more closely connected to their reaction than the situation itself' (*Wright and Beck, 1983*). The therapies consist of different components such as self-monitoring, goal setting and the development of alternative behaviours (*Castelnuovo et al., 2017*) by using thought records and behavioural experiences (*Werrij et al., 2009*).

A number of studies have examined the effects of a psychological intervention in adult women with PCOS. The objective of this systematic review is to examine the different types of CBT interventions and to determine the effects of these on depression scores in adult women with PCOS by performing a meta-analysis.

METHODS

Literature search

Relevant studies published up to July 2020 were identified by a biomedical information specialist in EMBASE, Medline (Ovid), Web of Science, Cochrane CENTRAL, PsycINFO (Ovid) and Google Scholar, using the following search terms: 'PCOS' and 'depression' or 'mood disorder' or 'psychology', or 'psychological wellbeing', or 'life satisfaction', or 'psychological aspect', or 'behaviour disorder', or 'eating disorder', or 'quality of life', or 'stress', or 'anxiety disorder'. Detailed information on the search strategy is listed in the Supplementary Information. Hand searches and reference checks were also performed. No language restriction was applied.

Study selection

After identifying and excluding duplicate studies, studies were screened based on the title and abstract to select studies that potentially met the inclusion criteria. The inclusion criteria for studies were: (i) adult women with PCOS; (ii) participants received CBT; (iii) the primary or secondary outcome measure was depression; (iv) validated psychological questionnaires were used to measure depression; (v) an intervention and control group were included; (vi) reported sufficient data to estimate Cohen's *d* effect sizes (mean, SD or SE and number of participants in each arm) in both intervention and care as usual; and (vi) were written in English. If necessary, the authors were contacted for additional data. Studies were excluded if (i) the intervention was conducted in adolescents with PCOS; (ii) if the intervention did not use any form of CBT; (iii) if the effect size could not be calculated. Two reviewers independently screened all studies (GJ and AB) on title and abstract using COVIDENCE (https://www.covidence.org/). There was no disagreement over eligibility of studies. The same authors performed the full-text screening to determine the final selection. This study was registered in PROSPERO (https://www.crd.york.ac.uk/prospero/) under registration number CRD42020173513.

Data extraction and quality assessment

For the systematic review, the PRISMA guidelines were followed (*Page et al., 2021*). For the interventions in the meta-analysis, the revised Cochrane risk of bias tool for randomized trials (RoB 2.0) was used (*McGuinness and Higgins, 2021*) to assess the risk of bias in five domains: randomization processes, deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result. Studies were categorized as 'low', 'some concerns' or 'high' risk of bias in all five domains. The overall quality of the study was categorized as 'low' risk of bias when all the domains scored low, 'some concerns' when at least one domain was scored as some concerns, and 'high' when at least one domain scored as high or more than two domains as some concerns. The data extraction and quality assessment were performed by two reviewers (GJ and AB) using a standardized extraction form. An independent reviewer (CZ) assessed the quality of the study by *Jiskoot et al. (2020)*.

Data syntheses and analysis

This meta-analysis was performed using the R package 'meta' (*Balduzzi et al., 2019*). To compare the different questionnaires for depression an effect size (using Cohen's *d*) was calculated for each study to compare the difference between baseline and post-intervention scores in both the intervention and control condition. A Cohen's *d* effect size of 0.2 is considered as small, 0.5 as medium and above 0.8 as large. This means that if the means of two groups do not vary by 0.2 SD or more, the difference is meaningless, even if it is statistically significant (*Cohen, 1988*). Effect sizes were calculated as the mean difference between depression scores for the intervention compared with care as usual divided by the pooled SD; negative effect sizes reflected deficits compared with care as usual. Subsequently, for each test, effect sizes were weighted using the inverse variance method within a random-effects model and pooled across all studies with available data. For one study, 0.001 was added to the average score to circumvent an exact difference of 0, to allow statistical analysis. In addition, a sensitivity analysis was conducted excluding papers with a high risk of bias based on the RoB 2.0 outcomes. A second sensitivity analysis was conducted excluding lifestyle intervention studies, as these studies included some form of dietary advice and exercise as part of their intervention, possibly leading to weight loss and confounding the observed change in depression scale. As a result of the various questionnaires to measure depression and the diversity in treatment, heterogeneity between studies was assumed. Heterogeneity between studies included in the meta-analysis was further evaluated using the I^2 statistic, which assesses the appropriateness of pooling the individual study results. A random-effects model was used in cases of considerable heterogeneity (I^2 >70%). Funnel plot asymmetry was not assessed because no more than 10 studies were included (*Sterne et al., 2011*).

A minimum clinically important difference (MCID) has been established for each questionnaire used, which is defined as the smallest change that patients and clinicians perceive as essential (*Jaeschke et al., 1989*). For the CES-D questionnaire a reduction of 11 points was defined as the MCID irrespective of age and gender (*Haase et al., 2021*). For the DASS, a reduction of 5.4 (29%) was found in a study in patients with chronic obstructive pulmonary disease (*Yohannes et al., 2019*). A reduction of between 3.86 and 6.15 points was calculated as the MCID for an outpatient sample (*Ronk et al., 2013*). For the BDI-II, a difference of ≥3 BDI-II points was suggested for normal depression (*National Collaborating Centre for Mental Health, 2010*). Another study estimated a MCID of 17.5% reduction from baseline (*Button et al., 2015*). No MCID is available for the Psychological General Well-Being Index (PGWBI).

RESULTS

Search results

The search strategy identified a total of 4854 articles, including duplicates and articles that had no relevance to the primary research questions (FIGURE 1). After removing these articles a total of 3105 were available. After review of abstracts, 46 articles were selected. Among them, eight articles appeared to be potentially appropriate for the systematic review. TABLE 1 summarizes the study design, PCOS diagnostic criteria, inclusion criteria, outcome measures, description of the intervention, intervention intensity and duration of the studies. For the meta-analysis, two authors were contacted by email to obtain additional depression data that were not reported in the original papers (*Cooney et al., 2018*; *Oberg et al., 2020*).

Description of studies

Of the eight studies, five studies were RCT that investigated the effects of an intervention on depression and were published between 2012 and 2020 in Europe (n = 3) (*Jiskoot et al., 2020*; *Oberg et al., 2020*; *Stefanaki et al., 2015*), North America (n = 1) (*Cooney et al., 2018*) or Asia (n = 1) (*Abdollahi et al., 2018*). One study was a qualitative study (*Roessler et al., 2012*), one was a clinical trial (*Ramazanzadeh et al., 2019*) and one a case report (*Correa et al., 2015*). These three studies could not contribute to the meta-analysis (*Correa et al., 2015*; *Ramazanzadeh et al., 2019*; *Roessler et al., 2012*). The number of participants in the included trials ranged from 15 to 183. Most trials assessed the effect of a psychological intervention (*Abdollahi et al., 2018*; *Stefanaki et al., 2015*), lifestyle intervention for weight loss (*Cooney et al., 2018*; *Jiskoot et al., 2020*; *Oberg et al., 2020*) or combined a pharmacological and psychological approach (*Ramazanzadeh et al., 2019*). Most trials investigated the effect of the intervention on symptoms of depression as primary outcome (*Abdollahi et al., 2018*; *Correa et al., 2015*; *Ramazanzadeh et al., 2019*; *Stefanaki et al., 2015*), and three trials assessed depression as a secondary outcome (*Cooney et al., 2018*; *Jiskoot et al., 2020*; *Oberg et al., 2020*). The study by *Roessler et al. (2012)* examined the psychological and communicative processes during group counselling in a qualitative study. Of the eight studies, three studies used the Rotterdam criteria for PCOS diagnosis (*Rotterdam ESHRE/ASRM-Sponsored PCOS Consensus Workshop Group, 2004*), one used the former National Institutes of Health (NIH) criteria (*Zawadski and Dunaif, 1992*) and in the remaining four studies the criteria were not specified.

Interventions and assessments

The duration of treatment in the included trials ranged from 8 to 52 weeks and involved between 8 and 20 sessions that lasted between 30 and 150 min per session. Of the eight included studies in the systematic review, four studies were group-based and four studies involved individual face-to-face sessions. Most trials examined some form of CBT: counselling based on the PCOS workbook (*Correa et al., 2015*), counselling based on a CBT approach (*Abdollahi et al., 2018*), brief CBT (*Cooney et al., 2018*), psychodynamic group therapy (*Roessler et al., 2012*), meetings with a lifestyle coach (*Oberg et al., 2020*) or structured CBT embedded in a group-based lifestyle programme (*Jiskoot et al., 2020*). One trial examined the effects of daily

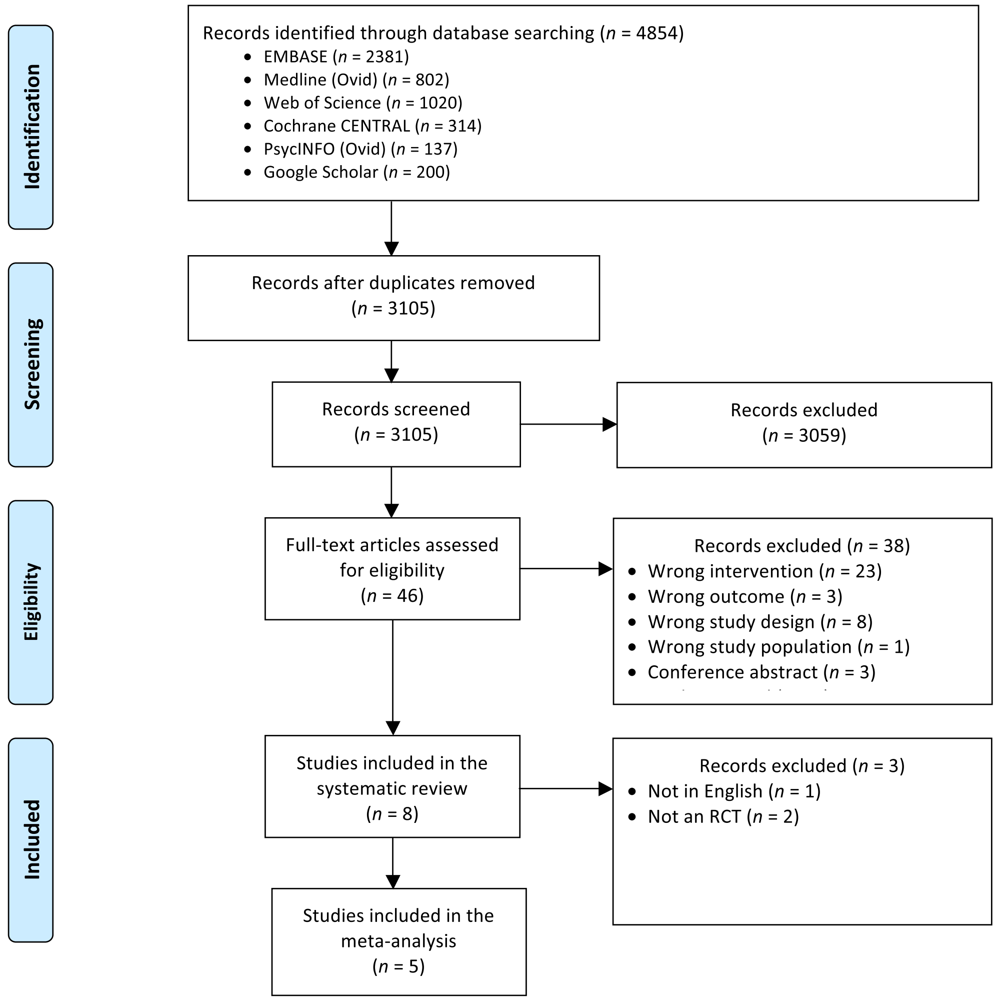

FIGURE 1 PRISMA flow diagram showing the selection of studies for inclusion.

mindfulness practice or mindfulness stress management (Stefanaki et al., 2015). The intervention that combined a pharmacological and psychological approach examined the use of fluoxetine alone or 12 weeks of supportive psychotherapy alone (Ramazanzadeh et al., 2019). In two interventions, care as usual consisted of no intervention (Abdollahi et al., 2018; Stefanaki et al., 2015). In three studies, the care as usual consisted of a minimal intervention without CBT (Cooney et al., 2018; Jiskoot et al., 2020; Oberg et al., 2020). One study consisted of a crossover design in which all participants received the same intervention in a different order (Roessler et al., 2012). Five different instruments were used to measure depression. The Beck Depression Inventory-II (BDI-II) was used most frequently. Other questionnaires to measure depression were the Depression Anxiety Stress Scale (DASS-21), the Center for Epidemiologic Studies Depression Scale (CES-D) and the Psychological General Well-Being Index (PGWBI) (TABLE 1).

Quality assessment

The RoB 2.0 tool was used for quality assessment of the RCT (FIGURE 2). Among the five included studies, three studies (Cooney et al., 2018; Jiskoot et al., 2020; Oberg et al., 2020) were classified as 'some concerns'. All five papers were classified as 'some concerns' for the second and fifth domain, 'deviations from intended interventions', and 'selection of reported results', in which it is stated that the authors should mention whether participants received the intended intervention, whether there was protocol deviation or whether a subset of data was used. These domains were not described in any of the papers. The study by Stefanaki et al. (2015) was classified as high risk of bias due to concerns over three domains. This study was classified as 'some concerns' for the domain 'missing outcome data' because the reasons for drop-out were not mentioned. The risk of bias of the study by Abdollahi et al. (2018) was classified as high for the domains 'randomization process' and 'bias in selection of the reported results' because the randomization process was not explained and no treatment description was mentioned.

Meta-analysis

For the meta-analysis, Cohen's *d* effect sizes were calculated based on five published articles (TABLE 2). A total of 136 participants in the intervention arm and 136 participants in the care as usual arm were included (Abdollahi et al., 2018; Cooney et al., 2018; Jiskoot et al., 2020; Oberg et al., 2020; Stefanaki et al., 2015). In the random fixed-effects model (FIGURE 3), four studies showed a positive Cohen's *d* and one study found no effect. An overall Cohen's *d* effect size of 1.02 (95% confidence interval [CI] 0.02–2.02) was found. This meta-

TABLE 1 CHARACTERISTICS OF STUDIES INCLUDED IN THE SYSTEMATIC REVIEW

First author, year of publication	Title	Journal, country of publication	Study design	PCOS criteria	Inclusion criteria	Outcomes	Intervention	Care as usual
Oberg et al., 2020	Psychological well-being and personality in relation to weight loss following behavioural modification intervention in obese women with polycystic ovary syndrome: a randomized controlled trial	European Journal of Endocrinology, Sweden	RCT	Rotterdam	BMI ≥27 kg/m^2 Between 18 and 40 years	Psychological General Well-Being Index (PGWBI) Personality traits (SSP)	4 months of weekly meetings (face-to-face) in small groups with lifestyle coach with monthly coaching sessions.	Received oral and written information about a healthy lifestyle from a midwife.
Cooney et al., 2018	Cognitive-behavioural therapy improves weight loss and quality of life in women with polycystic ovary syndrome: a pilot randomized clinical trial	Fertility and Sterility, USA	Pilot RCT	NIH	CES-D score ≥14 BMI 27–50 kg/m^2	Depression (CES-D)	16 weekly individual face-to-face sessions (30 min) of nutritional advice/exercise + 8 weeks brief CBT (30 min).	16 weekly individual face-to-face sessions (30 min) of nutritional advice/exercise.
Stefanaki et al., 2015	Impact of a mindfulness stress management program on stress, anxiety, depression and quality of life in women with polycystic ovary syndrome: a randomized controlled trial	The International Journal on the Biology of Stress, Greece	Pilot RCT	Rotterdam	Between 15 and 40 years	Depression (DASS-21)	8 weekly (30 min individual face-to-face) sessions of daily mindfulness stress management programme by audio and guided instructions.	No intervention, only questionnaires.
Ramazanzadeh et al., 2019	The effect of psychiatric interventions on depression and fertility in infertile patients with polycystic ovarian disease	Journal of Research in Medical and Dental Science, Iran	Clinical trial	Unknown	Between 20 and 35 years	Depression (BDI-II)	Fluoxetine 20–60 mg/day or supportive psychotherapy for 12 weeks (1–2 times per week individual face-to-face) or combination of both methods in infertile women with PCOS.	Fluoxetine 20–60 mg/day or supportive psychotherapy for 12 weeks (1–2 times per week individual face-to-face) or combination of both methods in infertile women without PCOS diagnosis.
Abdollahi et al., 2018	The effect of cognitive behavioural therapy on depression and obesity in women with polycystic ovarian syndrome: a randomized controlled clinical trial	Iranian Red Crescent Medical Journal, Iran	RCT	Unknown	Between 18 and 35 years Completed a secondary school education	Depression (BDI-II)	8 weekly sessions (45–60 min) cognitive behavioural approach in groups (8 to max. 10 participants).	No intervention, only questionnaires.
Jiskoot et al., 2020	Long-term effects of a three-component lifestyle intervention on emotional well-being in women with polycystic ovary syndrome (PCOS): a secondary analysis of a randomized controlled trial	PLOS One, the Netherlands	RCT	Rotterdam	BMI ≥25 kg/m^2 Between 18 and 38 years	Depression (BDI-II) Self-esteem (Rosenberg) Body image (fear of negative appearance)	One year of 20 group CBT sessions (90 min) with psychologist and dietician and physical therapy sessions (60 min).	One year of five individual face-to-face meetings with treating physician. Participants were encouraged to lose weight through publicly available services.
Correa et al., 2015	A case report demonstrating the efficacy of a comprehensive cognitive-behavioural therapy approach for treating anxiety, depression, and problematic eating in polycystic ovarian syndrome	Archive of Women's Mental Health, USA	Case report	Unknown	NA	Depression (BDI-II) Anxiety (BAI) General psychosocial functioning (OQ) Personality characteristics (PAI) Eating disorder (EDEQ)	11 individual face-to-face sessions primarily focused on material presented in *The PCOS Workbook*	NA
Roessler et al., 2012	Supportive relationships – psychological effects of group counselling in women with polycystic ovary syndrome (PCOS)	Communication and Medicine, Denmark	Crossover trial	Unknown	BMI 25–40 kg/m^2 Premenopausal	The group counselling sessions were filmed, tape recorded and transcribed verbatim	8 weekly (90 min) face-to-face sessions of high-intensity aerobic exercise and 8 weekly (90 min) psychodynamic group therapy sessions (crossover).	8 weekly (90 min) psychodynamic group therapy sessions and 8 weekly (90 min) face-to-face sessions of high-intensity aerobic exercise (crossover).

BAI = Beck Anxiety Inventory; BDI-II = Beck Depression Inventory-II; BMI = body mass index; CBT = cognitive behavioural therapy; CES-D = Center for Epidemiologic Studies Depression Scale; DASS-21 = Depression Anxiety Stress Scale; EDEQ = Eating Disorder Examination Questionnaire; NA = not available; NIH = National Institutes of Health; OQ = Outcome Questionnaire-45.2; PAI = Personality Assessment Inventory; PCOS = polycystic ovary syndrome; RCT = randomized controlled trial.

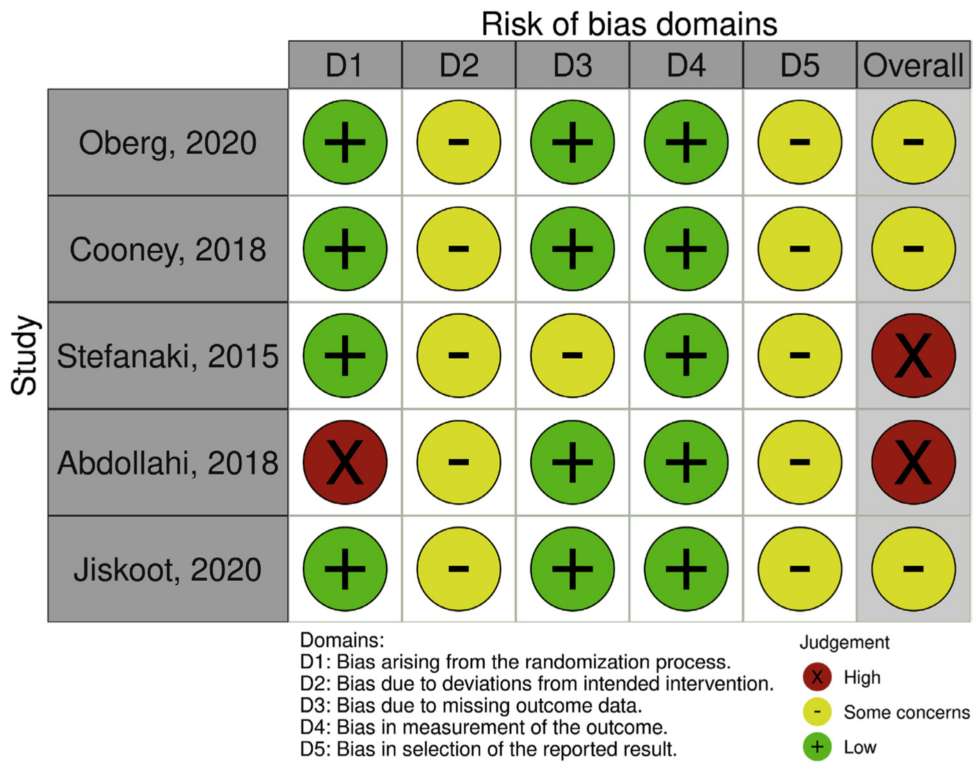

FIGURE 2 Results of the RoB 2.0 tool risk of bias assessment.

analysis showed a large effect size in favour of the intervention compared with care as usual. However, there is large between-study heterogeneity in intervention effect (I^2 = 88%; P < 0.001), suggesting that 88% of the variability in treatment effect estimates is due to differences (heterogeneity) between included studies and 12% might be due to chance. Therefore, two additional sensitivity analyses were performed. In the first sensitivity analysis one study was excluded (Abdollahi et al., 2018) due to a high risk of bias and unknown criteria for the diagnosis of PCOS (FIGURE 4). In this analysis, an overall Cohen's d of 0.66 (95% CI 0.26–1.06) was found in the fixed-effects model, which is considered a medium effect size in favour of the intervention. Exclusion of this study resulted in a lower I^2 statistic of 53%. In the second sensitivity analysis, three lifestyle interventions for weight loss were excluded (Cooney et al., 2018; Jiskoot et al., 2020; Oberg et al., 2020). In this analysis, an overall Cohen's d of 2.11 (95% CI 1.64–2.59) was found in the fixed-effects model, which is considered a large effect size in favour of the intervention. Exclusion of these three lifestyle interventions resulted in an I^2 statistic of 85% (FIGURE 5).

A MCID was found in three out of five studies; a MCID was achieved in the intervention groups and not in the control groups (TABLE 2) (Abdollahi et al., 2018; Jiskoot et al., 2020; Stefanaki et al., 2015).

DISCUSSION

The primary objective of this systematic review and meta-analysis was to summarize the current evidence for the effect of psychological interventions on depression scores in women with PCOS. In this systematic review of the literature, three RCT were found reporting on interventions for depression as primary outcome, while other RCT reported on depression as a secondary outcome measure in a lifestyle intervention. A large statistically significant effect size

TABLE 2 BASELINE AND POST-INTERVENTION SCORES FOR DEPRESSION

Study	Measurement	Intervention			Care as usual			MCID
		n	Mean baseline (SD)	Mean post (SD)	*n*	Mean baseline (SD)	Mean post (SD)	in symptoms
Oberg et al., 2020	PGWBI (DM)	34	10.00 (1.8)	10.00 (0.9)	34	10.00 (1.3)	10.00 (0.9)	Unknown
Cooney et al., 2018	CED-D	15	20.75 (8.4)	15.88 (11.8)	15	21.71 (4.9)	21.00 (6.8)	No
Stefanaki et al., 2015	DASS-21	23	18.00 (8.0)	4.34 (3.3)	23	18.10 (12.8)	17.20 (10.0)	Yes, in the intervention
Abdollahi et al., 2018	BDI-II	37	16.40 (0.6)	4.50 (3.9)	37	13.70 (5.7)	16.50 (8.6)	Yes, in the intervention
Jiskoot et al., 2020	BDI-II	27	16.07 (11.1)	11.59 (11.3)	27	8.75 (6.8)	8.27 (8.8)	Yes, in the intervention

BDI-II = Beck Depression Inventory-II; CES-D = Center for Epidemiologic Studies Depression Scale; DASS-21 = Depression Anxiety Stress Scale; MCID = minimum clinically important difference; PGWBI = Psychological General Well-Being Index.

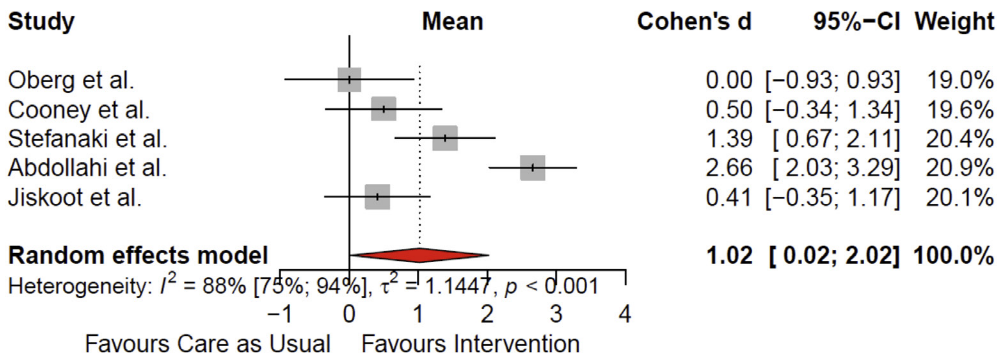

FIGURE 3 Overall forest plot for meta-analysis of psychological interventions compared with care as usual in women with PCOS.

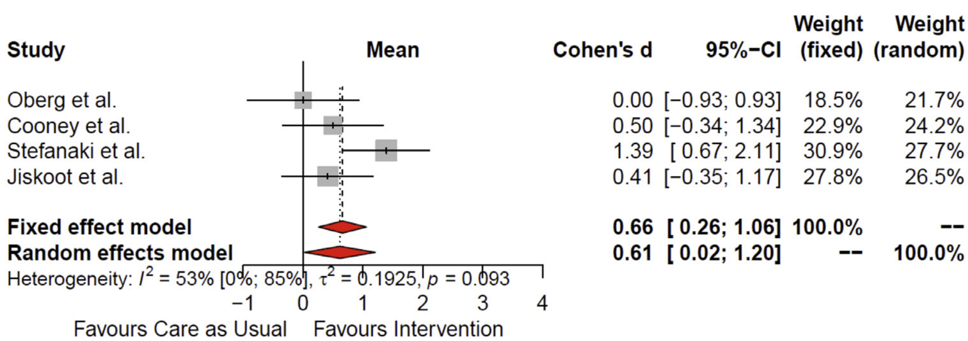

FIGURE 4 Forest plot based on sensitivity analysis with the study with high risk of bias excluded.

(d = 1.16) was found in favour of all the interventions compared with care as usual for depression in women with PCOS. Psychological interventions including CBT seem to be an effective form of therapy in decreasing depression scores in women with PCOS.

This is thought to be the first systematic review and meta-analysis that has focused on psychological interventions for depression in women with PCOS. CBT was used in almost all interventions that were included in this meta-analysis. CBT was defined as a psychological intervention in which restructuring of negative beliefs was one of the central components. However, most RCT also included other components such as homework assignments, lifestyle counselling, behavioural modification, exercise and mindfulness and used different protocols of CBT. Also, three of the studies were group-based interventions and the other two interventions were based on individual sessions. The variety of these components and formats differed widely. In the general population, a recent network meta-analysis revealed no differences between CBT protocols for depression. This implies that individual treatment, group, telephone and guided self-help CBT were all effective (Cuijpers et al., 2019), although others found that individual CBT was more effective than group-based CBT (Cuijpers et al., 2008). However, in the current meta-analysis a large variation in Cohen's d effect sizes was seen when comparing individual studies using different psychological interventions in women with PCOS. This result suggests that this difference could be explained by the CBT protocol, individual versus group treatment or length of treatment. Also, three of the included studies involved a lifestyle intervention with a great emphasis on weight loss. In these interventions, CBT is one of the components besides nutritional advice and exercise. Based on the current literature, it is known that a 5–10% weight loss can improve depression, and the reproductive and metabolic features of PCOS (Naderpoor et al., 2015; Teede et al., 2010). It was decided to include these lifestyle

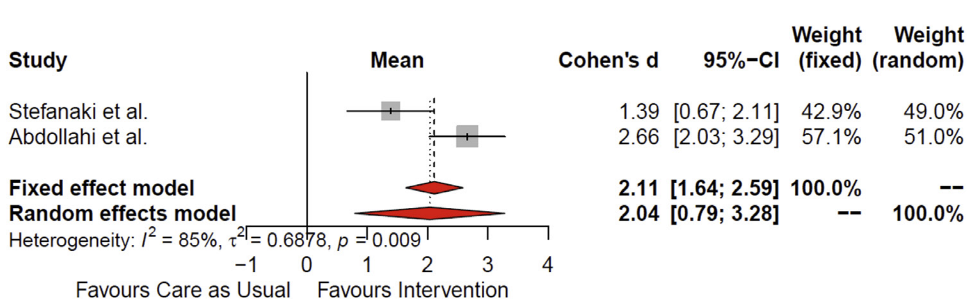

FIGURE 5 Forest plot based on sensitivity analysis with the three studies with lifestyle intervention for weight loss excluded.

interventions in this meta-analysis because it is possible that weight loss itself, more than a specific CBT protocol within the lifestyle interventions, was responsible for the changes in depression in women with PCOS. To assess this hypothesis, studies that incorporated dietary advice and exercise within their lifestyle interventions were excluded, leaving CBT as the main intervention. In this sensitivity analysis, an even larger effect size (d = 1.99) in favour of CBT compared with care as usual was found for depression in women with PCOS. Although a large effect was found for the intervention group compared with care as usual, it is believed that the differences in the interventions and CBT protocols may have contributed to the statistical heterogeneity and certainly to the clinical heterogeneity found in this meta-analysis.

Besides the concerns over heterogeneity, this meta-analysis has several limitations. First, it is based on a moderate number of individuals with PCOS within an even smaller number of studies. A small sample size can be a problem when calculating the degree of heterogeneity. Particularly when the number of patients and studies is low, the risk of overestimating the I^2 statistic is high (*Thorlund et al., 2011*; *von Hippel, 2015*). A second limitation of this meta-analysis relates to the diagnosis of PCOS; it was found that the criteria for PCOS diagnosis were not identical between studies. Most studies used the Rotterdam criteria and in several studies the criteria for PCOS were unknown. In a recent meta-analysis that examined mental health in women with PCOS, differences in diagnostic criteria were significantly associated with heterogeneity in results (*Yin et al., 2021*). Based on these limitations, the current findings should be interpreted with caution. Moreover, they again underpin the necessity of adopting the international guideline (*Teede et al., 2018a*) recommending the use of the Rotterdam criteria to diagnose PCOS.

Given the high prevalence and odds for depression in women with PCOS (*Cooney et al., 2017*; *Dokras et al., 2011*), the PCOS guideline recommends that all women should be screened for depression. Besides screening for depression, more research is needed to optimize psychological treatment options for women who suffer from depression and mood disorders. Therefore, larger studies should examine whether a standardized CBT protocol or pharmacological options are effective for depression in women with PCOS. To control for bias, it is suggested that future trials investigating the effect of psychological interventions on depression in women with PCOS should be carried out by collaborative research teams.

In conclusion, this systematic review and meta-analysis demonstrates that most psychological interventions applying CBT are effective in lowering depression scores in women with PCOS. A MCID was found in four out of five interventions. However, these results should be interpreted with caution due to methodological differences and the low quality of the studies. Therefore, more clinical trials are needed to assess whether CBT or pharmacological options are effective to improve depression in women with PCOS.

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DATA AVAILABILITY STATEMENT

The authors confirm that the data supporting the findings of this study are available within the article.

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